

Versatile Low-Cost Power Supply

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LM317 is a very popular, low-cost and useful adjustable series voltage regulator. The maximal power dissipation is up to 20W. The current-limiting function of LM317 is set to around 1.5A, and sometimes that is too high for the modern low-power equipment.

Circuit and working

Circuit diagram of the low-cost power supply with fixed and adjustable capabilities is given in Fig. 1. The mains power supply is applied to the primary step-down transformer X1 through connector CON1. The primary voltage rating of the transformer is 230V AC, while of the secondary is 22V AC. Its current rating is 1.5A.

Noise in the mains is suppressed with capacitor C1 and resistor R1. Bridge rectifier BR1 with current rating of 1A converts AC voltage into DC voltage. Filtering is done by

capacitors C2 through C7. Capacitors C8 through C11 are needed for stability of IC1 and IC2. Diodes D1 and D3 provide protection to IC1 and IC2 against reverse voltage.

IC1 is a current-limiting regulator with five output currents. The output current-limiting function is selected with switch S2 and resistors R2 through R6. Positions of the switch and the corresponding maximum output current are given below:

Switch S2 position	Output current (A)
1	1.00
2	0.50
3	0.35
4	0.22
5	0.10

IC2 works as a multi-level linear-voltage regulator to provide adjustable and fixed voltage outputs. Selection of adjustable and fixed voltages is done through switch S3. When switch S3 is in position 1, output in

the range 1.2V-16V is available at connectors CON2, CON3 and CON4. Adjustment of output voltage is done using potentiometer VR11.

When switch S3 is in position 3 and switch S4 is in one of the ten positions, a fixed voltage will be available at connectors CON2, CON3 and CON4. Selection of the fixed voltage in ten steps is done by switch S4. Trimming of the fixed voltage to a predetermined level is done by potentiometers VR1 through VR10. Also, we can replace some, or all, of the potmeters with appropriate fixed resistors.

Let us assume, you need to check the performance of an electronic equipment at different voltages, say, 1.2V, 2V, 3V, 4V, 5V and 5.5V. For this, switch S3 should be in position 3 and position of switch S4 should be changed from 1 to 6. Setting of VR1

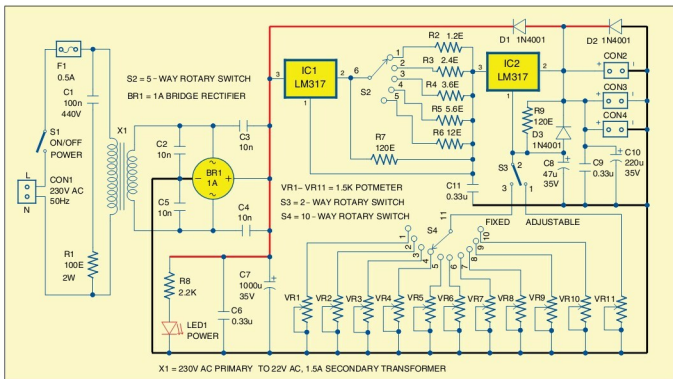


Fig. 1: Circuit diagram of the versatile power supply